

University of Verona

Department of Economics
Master's Degree in Economics and Data Analysis

Time Series and Forecasting

Forecasting Crude Oil Price Volatility: A Time Series Analysis of WTI Spot Prices (2000–2019)

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We begin by uploading the dataset in **MATLAB** using the `readtable()` function, which loads the CSV file containing monthly Spot Crude Oil Prices for West Texas Intermediate (WTI) from January 2000 to December 2019.

To stabilize the series and focus on relative movements, we compute the first differences of the price series and store them in a variable named `oilChanges`, which will be central to the analysis.

We then plot the resulting time series of monthly oil price changes:

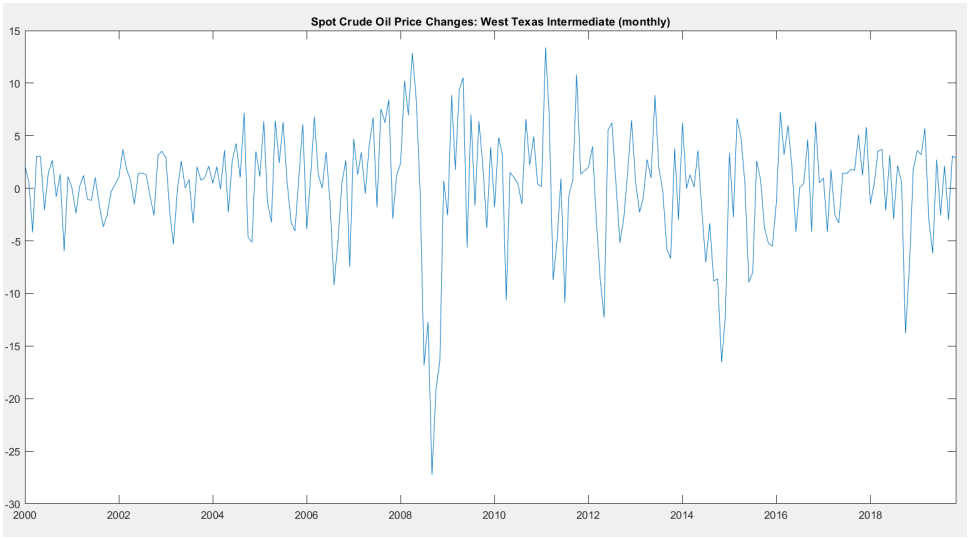


Figure 1: Monthly changes in WTI crude oil prices (2000–2019)

The series is visibly volatile, as expected from a commodity that is highly sensitive to geopolitical events, economic shocks, and supply-demand dynamics. Between 2000 and 2019, several major events significantly influenced the price of crude oil. Among them, the two most impactful are:

- **The 2008 Global Financial Crisis:** This event triggered a sharp decline in oil demand, reflected in a dramatic drop of -27.25 in oil price changes in September 2008 — the most extreme value observed in the sample.
- **The 2014–2016 Oil Price Collapse:** Triggered by OPEC's decision not to cut production despite rising U.S. shale output, this event led to a prolonged oversupply and a substantial price decline.

Given this context, we choose to set t^* at two-thirds of the sample, which provides a sufficiently large training set while preserving a meaningful out-of-sample window for evaluation.

Interestingly, this split includes the 2008 financial crisis in the training set and leaves the 2014–2016 oil price collapse in the test set, making it a particularly informative setup for assessing model performance under different market conditions.

Using the first sub-sample, we performed the in-sample analysis. We began by examining the sample autocorrelation function (ACF) and partial autocorrelation function (PACF), which are useful tools for identifying the underlying structure of a time series. The results are shown in Figure 2.

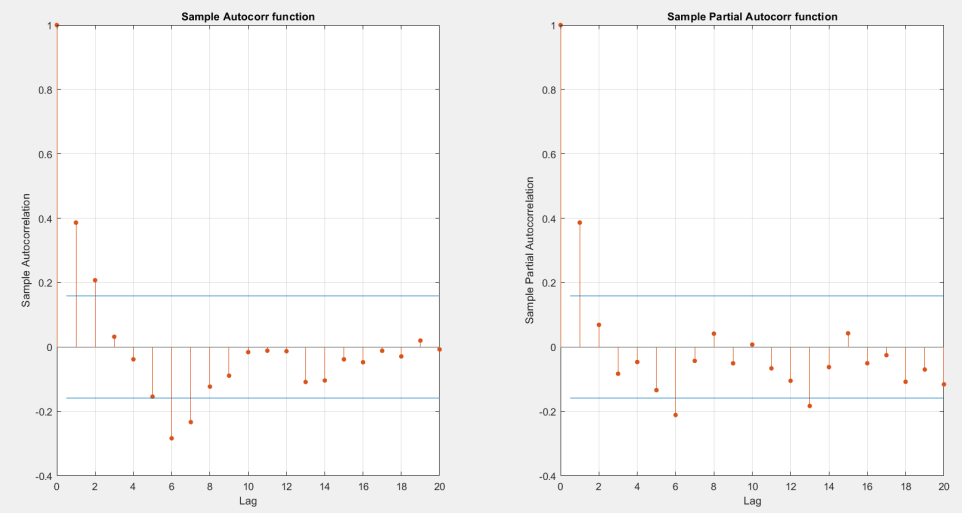


Figure 2: Sample ACF and PACF of the first sub-sample.

By looking at the ACF, we notice that it becomes zero after lag 2. This could suggest that the time series is generated by a moving average process of order 2 (MA(2)). From the PACF, we observe that it drops to zero after lag 1, indicating the possibility of an autoregressive process of order 1 (AR(1)). However, further tests need to be done to select the most appropriate model. At first, we selected several candidate models to describe our data and compared their performance using Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). These criteria introduce a penalty for adding parameters, helping to avoid the problem of overfitting when evaluating model fit. The candidate models we considered are: AR(1), AR(2), MA(1), MA(2), ARMA(1,1), ARMA(1,2), ARMA(2,2). The corresponding AIC and BIC values for each model are reported in Figure 3.

```
Model comparison (in-sample):
Model        AIC        BIC
AR(1)        979.604    988.811
AR(2)        980.852    993.128
MA(1)        985.575    994.782
MA(2)        980.295    992.571
ARMA(1,1)    981.081    993.357
ARMA(1,2)    982.140    997.484
ARMA(2,2)    968.729    987.142
```

Figure 3: AIC and BIC values for different candidate models.

Due to their respective construction, AIC tends to favour more complex models, whereas BIC rewards parsimony, often leading to divergent model suggestions. To select the best-performing model, we relied on AIC results, which are commonly preferred in forecasting contexts due to their emphasis on predictive accuracy.

Based on the AIC values, the models selected for further analysis are:

- ARMA(2,2)

- AR(1)
- MA(2)

The following step consisted in estimating the residuals (shocks) and checking whether they behave as the models assume, i.e., as strong white noise. The results are presented in the following figures:

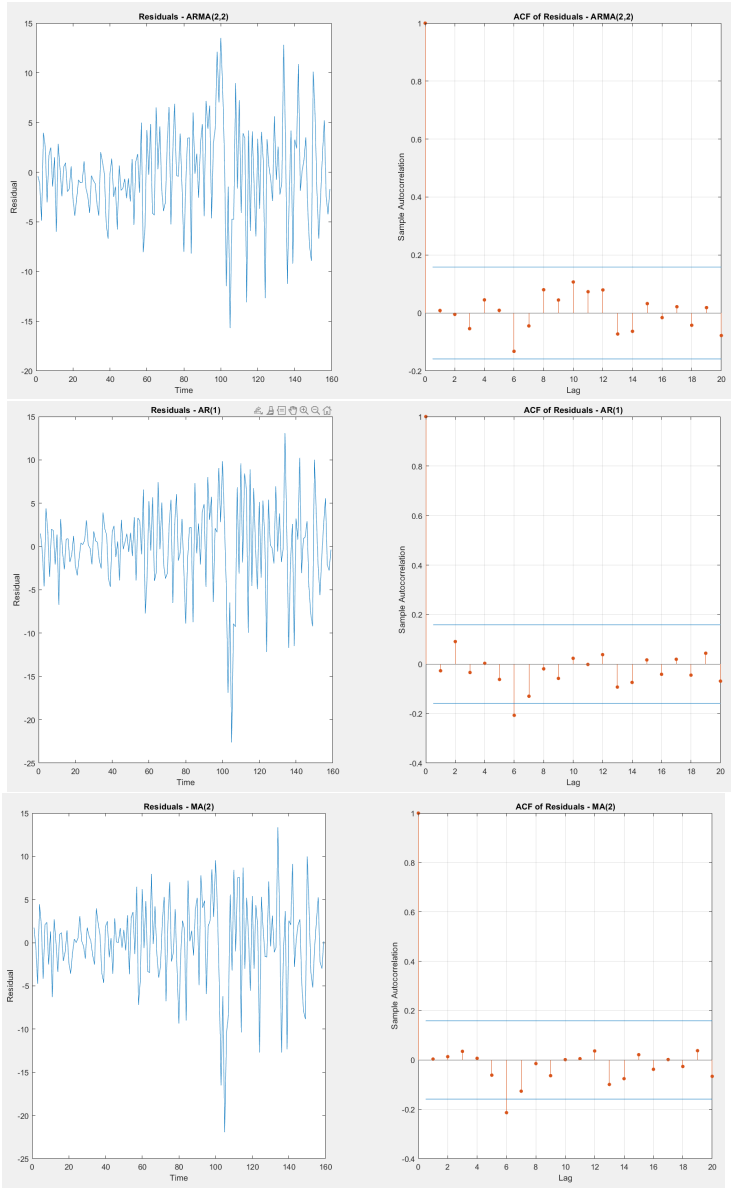


Figure 4: Residual diagnostics for ARMA(2,2), AR(1), and MA(2) models. Left panels show time plots of residuals; right panels show ACF of residuals.

In all three models, the left-side plots reveal increasing variance over time, indicating non-constant variance (i.e., non-homoskedasticity). While not ideal, this is not necessarily problematic as long as the residuals are not autocorrelated. This condition appears to be met in all models. Although some small autocorrelations are statistically significant in AR(1) and MA(2), they do not alter the overall absence of serial correlation.

In conclusion, residuals behave as weak white noise, not strong white noise as assumed by ARMA models, indicating that some volatility dynamics may not be fully captured—an acceptable shortcoming in the context of financial time series.

We then used the three selected models to perform in-sample forecasts for different time horizons (1, 2, and 6 months) and compared them with two benchmark methods: a flat forecast and an exponentially weighted moving average (EWMA). The results are shown below:

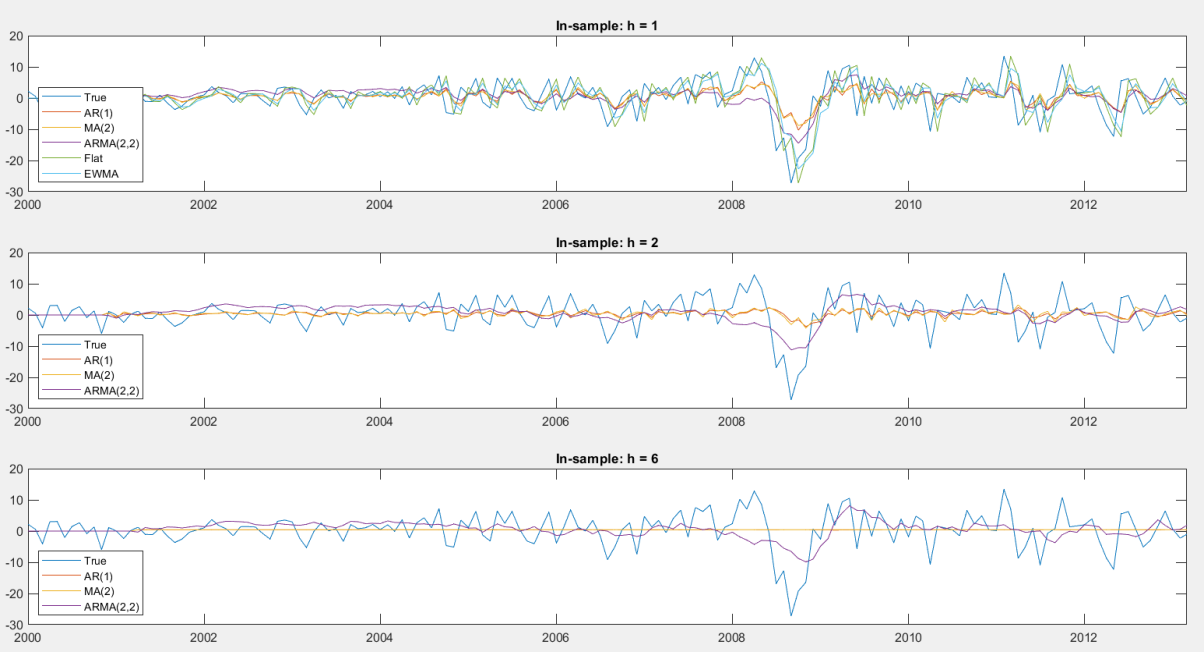


Figure 5: In-sample forecasts for 1, 2, and 6 months ahead.

From the graphs, it is evident that forecast accuracy declines as the forecast horizon increases. This is especially pronounced in the six-month forecast, where ARMA(2,2) is the only model that somewhat tracks the actual movements of oil prices.

Focusing on the one-month horizon, we computed the mean squared error (MSE) between predicted and actual values as a measure of forecast accuracy. A lower MSE indicates better predictive performance. The MSE-based ranking of the models is:

- ARMA(2,2)
- MA(2)
- AR(1)
- EWMA
- Flat forecast

```
In-sample MSE (h=1):
Model         MSE
AR(1)         28.033
MA(2)         27.708
ARMA(2,2)     25.171
Flat          40.223
EWMA          34.709
```

Figure 6: Mean Squared Error (MSE) for 1-month ahead forecast.

To conclude, based on the in-sample analysis, the ARMA(2,2) model emerges as the best-performing forecasting model for Spot Crude Oil prices.

Using the data from the second sub-sample, we computed the out-of-sample forecasts for horizons $h = 1, 2, 6$, using three ARMA-type time series models: AR(1), MA(2), and ARMA(2,2). For horizon $h = 1$, we also included two model-free approaches: a flat forecast and an exponentially weighted moving average (EWMA) forecast. The models were evaluated using the Mean Squared Error (MSE) on the out-of-sample period. The results for $h = 1$ are:

```
Out-of-sample MSE (h=1):
Model         MSE
AR(1)         22.102
MA(2)         21.708
ARMA(2,2)     92.330
Flat          32.915
EWMA          28.126
```

Figure 7: Out-of-sample MSEs by model for $h = 1$

Model ranking based on MSE (from best to worst): MA(2), AR(1), EWMA, Flat forecast, ARMA(2,2)

These results suggest that simpler models, such as MA(2) and AR(1), outperform more complex ones like ARMA(2,2) in out-of-sample forecasting. The ARMA(2,2) model exhibits the worst performance, possibly due to overfitting. The flat forecast and EWMA offer better results but are also outperformed by the AR and MA models.

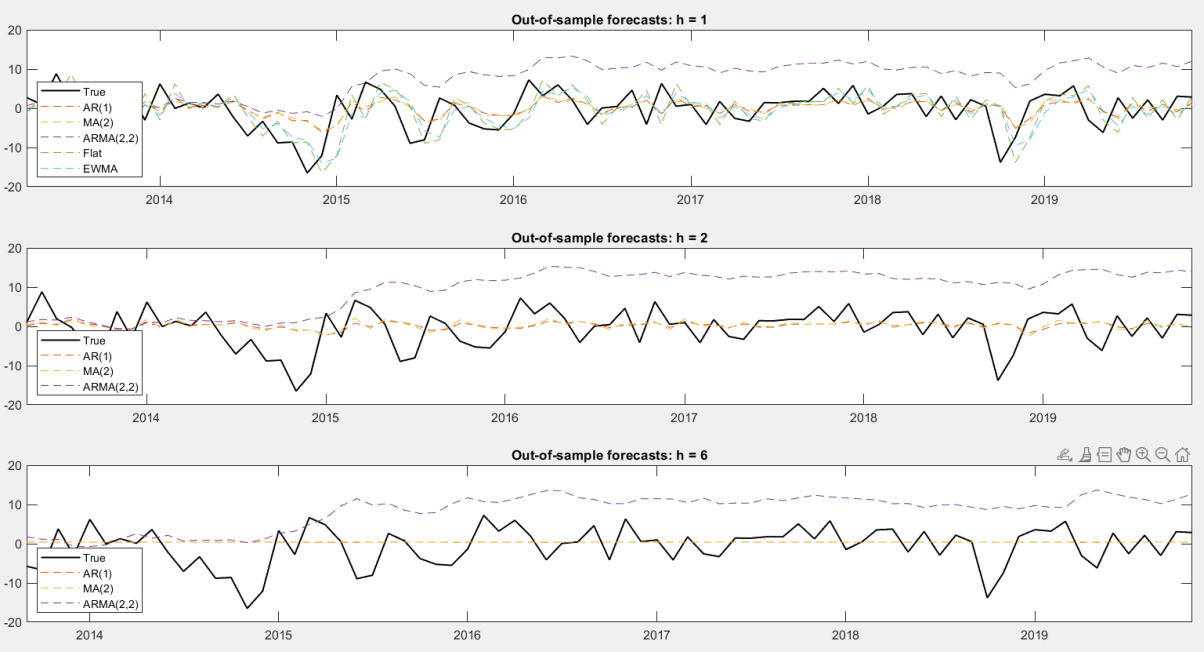


Figure 8: Out-of-sample forecasts for $h = 1, 2, 6$

The figure shows the out-of-sample forecasts for $h = 1, 2, 6$. As expected, forecast uncertainty increases with the horizon. The ARMA(2,2) forecasts consistently deviate more from the actual values across all horizons, confirming its inferior out-of-sample performance.

To assess whether these forecast differences are statistically significant, we used the Diebold-Mariano test. The DM statistics for $h = 1$ are:

```
----- Diebold-Mariano Test (h = 1) -----
ARMA(2,2) vs AR(1):
    stat: 7.7911
    pval: 6.6613e-15

ARMA(2,2) vs MA(2):
    stat: 7.8957
    pval: 2.8866e-15

ARMA(2,2) vs EWMA:
    stat: 6.3642
    pval: 1.9637e-10

ARMA(2,2) vs Flat:
    stat: 5.7835
    pval: 7.3174e-09

MA(2) vs AR(1):
    stat: -0.7837
    pval: 0.4332

EWMA vs AR(1):
    stat: 1.7600
    pval: 0.0784

EWMA vs Flat:
    stat: -3.2913
    pval: 9.9739e-04
```

Figure 9: Diebold–Mariano test statistics and p-values for $h = 1$

The high positive Diebold–Mariano (DM) statistics for ARMA(2,2) compared to the

other models indicate that ARMA(2,2) performs significantly worse. The DM statistic between MA(2) and AR(1) is small and negative (−0.7784), suggesting no significant difference in their predictive accuracy. The EWMA vs AR(1) statistic (1.7484) might indicate a marginally significant improvement of EWMA over AR(1), at the 10% significance level. Finally, the strongly negative value for EWMA vs Flat (−3.2738) indicates that EWMA significantly outperforms the Flat forecast.

Both the MSE ranking and the DM test support the conclusion that simpler models (MA(2), AR(1)) provide the best forecasting performance, while ARMA(2,2) performs significantly worse. These results underline the importance of parsimony in time series forecasting, as overly complex models tend to overfit and generalize poorly out-of-sample.

The overall findings of our out-of-sample forecasting analysis highlight the superior performance of simpler and more parsimonious models such as MA(2) and AR(1), both in terms of forecast accuracy (MSE) and statistical comparison (DM test). In contrast, the more complex ARMA(2,2) model exhibits poor performance and a tendency to overfit.

One important factor that may have influenced these results is the choice of the split point t^*, which was set relatively late in the time series. As a result, the training dataset includes the period of a major economic crisis, which introduces substantial volatility and structural changes into the estimation window.

This crisis-driven instability may have distorted the parameter estimates of more complex models like ARMA(2,2), making them less robust and poorly suited for generalization. Simpler models, by contrast, are more resilient to structural breaks and tend to perform better when facing volatile or nonstationary environments.

Thus, the combination of a late t^* and the inclusion of a turbulent period in the training data likely contributed to the inferior performance of complex models, reinforcing the idea that model simplicity can be a strength in the presence of economic uncertainty.

To evaluate the robustness of both in-sample and out-of-sample forecast results, we re-estimated the models using a different time-series partition, setting the split point t^* to 50% of the sample instead of the initial 2/3 value. This allows a direct comparison between the two specifications and reveals whether the choice of t^* significantly affects model performance and ranking.

Based on the first half of the sample, and by focusing on AIC criteria, we identify, as before, ARMA(2,2) as the best-fitting model, followed by MA(2) and AR(1). Compared to the previous setup, AR(1) becomes relatively less competitive, especially in terms of in-sample forecasting. However, ARMA(2,2)'s BIC is higher than that of MA(2) and AR(1), reflecting the penalty for model complexity.

```
Model comparison (in-sample):
Model        AIC        BIC
AR(1)        732.113    740.476
MA(2)        730.063    741.213
ARMA(2,2)    725.374    742.099
```

Figure 10: AIC and BIC model selection with 50% split

Then we plotted the updated autocorrelation and partial autocorrelation plots. We noticed that stronger higher-lag dependencies are visible compared to the previous ACF

structure. This suggests a richer short-term dynamic, encouraging the use of more flexible ARMA models.

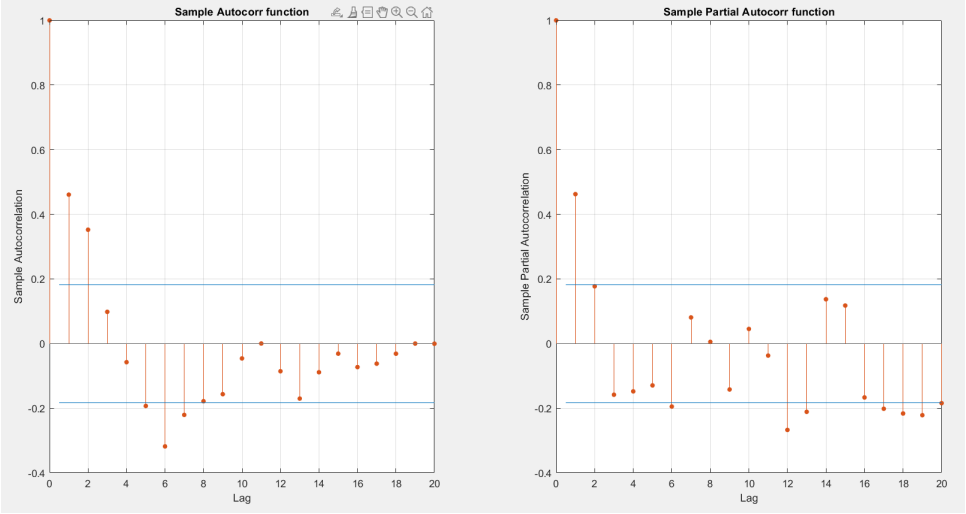


Figure 11: Updated ACF and PACF plots

In-sample forecasts for horizons $h = 1$, $h = 2$, and $h = 6$ are shown in the following figure.

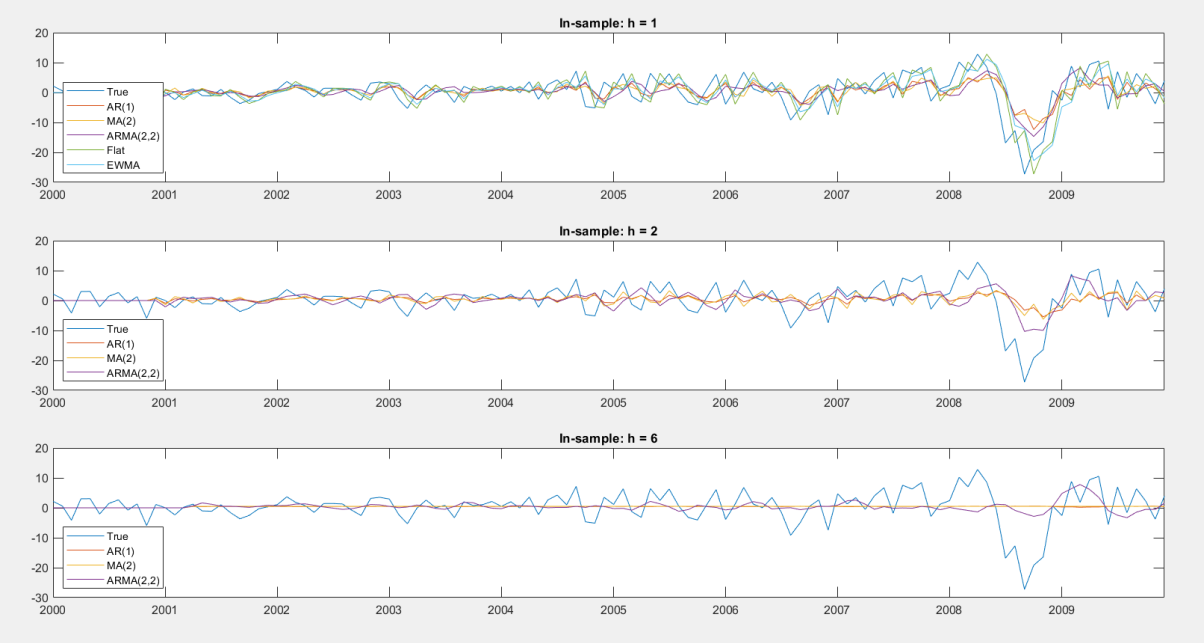


Figure 12: In-sample forecasts for $h = 1, 2, 6$

For $h = 1$, ARMA(2,2) clearly provides the best fit to the observed series, followed by MA(2) and AR(1). This ranking is confirmed by the in-sample MSEs:

In-sample MSE (h=1):

Model	MSE
AR(1)	26.416
MA(2)	25.259
ARMA(2,2)	23.675
Flat	35.825
EWMA	30.887

Figure 13: In-sample Mean Squared Errors by model

These results suggest that more flexible models perform better in-sample, especially when the training window is shorter and concentrated on recent dynamics.

The following figure displays the out-of-sample forecasts from 2010 to 2019:

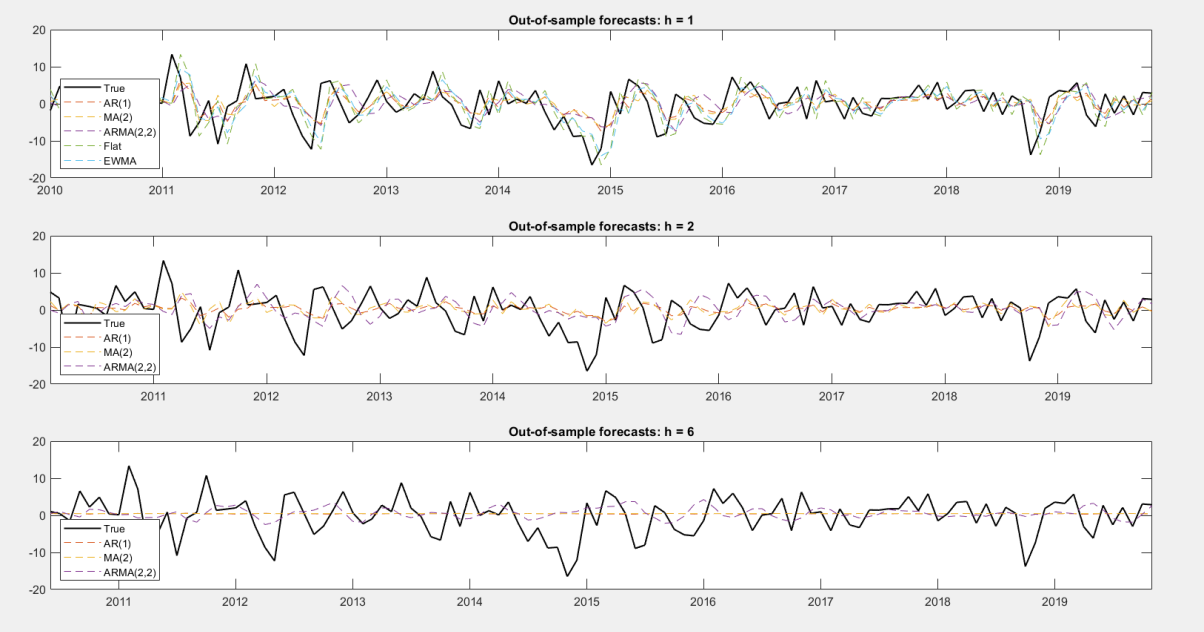


Figure 14: Out-of-sample forecasts (2010–2019)

Interestingly, the performance ranking changes compared to the in-sample results. AR(1) provides the lowest MSE out-of-sample, followed by MA(2), ARMA(2,2), EWMA, and FLAT:

```
Out-of-sample MSE (h=1):
Model         MSE
AR(1)         25.974
MA(2)         26.627
ARMA(2,2)     30.011
Flat          39.375
EWMA          33.817
```

Figure 15: Out-of-sample MSEs by model

This reordering illustrates a common phenomenon: more complex models may overfit the training data, especially in small samples, and thus generalize poorly. Simpler models, like AR(1), although suboptimal in-sample, offer greater out-of-sample stability.

The change in model rankings when shifting from $t^* = 2/3$ to $t^* = 1/2$ demonstrates that results are not fully robust to the choice of the sub-sample split. While ARMA(2,2) consistently dominates in-sample, its out-of-sample performance declines substantially, particularly when trained on earlier parts of the sample.

Some possible explanations could be the presence of structural breaks: splitting the sample earlier means that the estimation set includes fewer crisis periods (e.g., 2008 or 2014–15), making the estimated model less responsive to volatility clustering in later periods. Another relevant factor is overfitting risk: the ARMA(2,2) model captures short-term dependencies well but may overfit when estimation windows are smaller.

Finally, model parsimony plays a role: the AR(1) model, despite being simpler, proves more robust and generalizable in out-of-sample forecasts.